

SOFTWARE ENGINEERING - PROJECT PROPOSAL

Thikana

A Smart Real Estate and Rental Management Platform

A complete digital real estate ecosystem for Bangladesh

Prepared by

Full Stack Alchemists Team

Name	Role
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Anusua Biswas Nidhi	QA Tester
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Team: Full Stack Alchemists

Submitted to

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1. Team Formation

Thikana is being developed by the Full Stack Alchemists Team, a student group from the Department of Computer Science and Engineering, University of Barishal. The team is structured around clear responsibilities so that planning, development, presentation, testing, and version control can move forward without unnecessary overlap.

#	Name	Role and Responsibility
01	Arafat Rahaman	Team Lead: coordinates the team, supports decision-making, monitors progress, and keeps the project aligned with academic requirements.
02	Abdur Rahman	Project Manager and Backend Developer: manages planning, backend logic, APIs, authentication, verification flow, and server-side coordination.
03	Rezoan Nafi	Presentation Lead: prepares presentation structure, slide flow, storytelling, and final project communication.
04	Anusua Biswas Nidhi	QA Tester: checks feature quality, tests workflows, reports bugs, and validates user-side scenarios.
05	Raufe Hasan Ruhaan	Git Controller: maintains repository discipline, branch control, version history, and code collaboration.

Table 1: Full Stack Alchemists - Member Roles

2. SWOT Analysis of Team

Before development, the team considered its strengths, weaknesses, opportunities, and threats. This analysis helps the team understand what can support the project and what risks need to be managed during implementation.

Strengths	Weaknesses
Clear division of work among team members. Strong academic motivation to solve a real local problem. Practical understanding of web-based systems, dashboards, and database-driven applications. Project scope includes both technical and business value.	Advanced AI and 360-degree features may be challenging within a short student timeline. Real NID verification and payment gateway integration may require external approval. Large-scale traffic and real property data are difficult to test in an academic prototype. Limited time for real market validation.
Opportunities	Threats
Bangladesh has a growing need for reliable digital property solutions. Students, job holders, families, and owners all face real rental and property-search problems. Free-tier tools such as Firebase, Figma, and Google Maps API can reduce initial cost. The idea can grow into a commercial platform after the academic version.	Existing brokers and informal property networks may resist platform adoption. Fake listings and fraud attempts can damage trust if moderation is weak. API costs may increase after free-tier limits. Legal, privacy, and verification issues may affect future deployment.

Table 2: SWOT Analysis - Full Stack Alchemists

3. Idea Generation

The team selected the real estate and rental management domain after observing that property searching in Bangladesh is still highly fragmented. People often rely on Facebook groups, posters, brokers, or personal contacts. These methods are useful in some cases, but they are not organized, verified, or efficient enough for a modern housing market.

3.1 Domain and Problem Identification

The selected domain is a combined real estate marketplace and rental management platform. The core problem is that users must use different channels for house rent, flat rent, bachelor housing, sublet, property buying, property selling, and agency communication. This creates confusion for tenants and buyers, while owners lose time because they cannot manage listings, enquiries, and trust-building from a single system.

3.2 Why Thikana Was Selected

- **Feasibility:** A working prototype can be built with common web technologies such as React.js, Node.js, MySQL, and Firebase.
- **Impact:** Housing is a basic need, and improving property search saves time, money, and effort for many users.
- **Innovation:** Features like area intelligence, 360-degree virtual tours, AI recommendations, and tenant verification make the project stronger than a simple TO-LET listing website.
- **Scope:** The system can start with a rental MVP and later expand into buy/sell, agency management, analytics, and payment integration.
- **Learning Value:** The project covers frontend, backend, database, authentication, maps, payment, analytics, and AI-based features.

4. Introduction and Overview

Thikana, meaning "address" in Bengali, is a web-based smart real estate and rental management platform designed for Bangladesh. It connects tenants, buyers, property owners, and real estate agencies inside one digital ecosystem. The system supports house rent, flat rent, bachelor housing, sublet, property buy/sell, and property management.

The platform is designed as more than a listing website. It helps users search properties using filters, view detailed images, explore maps, check nearby facilities, communicate with owners, save favorites, and complete essential steps digitally. Owners can post listings, upload photos and floor plans, respond to enquiries, view analytics, and manage availability. Admins can verify users, moderate listings, handle reports, and monitor the whole platform.

The expected result is a trusted, transparent, and faster property-search experience. Instead of visiting multiple unsuitable houses or depending fully on brokers, users can make better decisions from verified information, location intelligence, and virtual previews.

5. Problem Statement

The current property search and real estate process in Bangladesh creates difficulties for tenants, buyers, owners, and agencies. Most information is scattered across social media, brokers, phone calls, or physical visits. Users often cannot confirm whether a listing is real, whether the price is fair, or whether the location is suitable before spending time and money visiting it.

Problem	Description
Fake Listings	Fraudulent or outdated posts mislead users and waste valuable time.
Broker Harassment	Middlemen may demand excessive or unclear commissions.
Time Wastage	Users visit multiple unsuitable properties due to lack of photos, maps, floor plans, and reliable details.
Poor Location Intelligence	There is often no organized information about nearby schools, hospitals, markets, transport, traffic, safety, or internet availability.
Trust Deficit	Many platforms do not verify tenants, owners, agencies, or property documents.
No Dispute Resolution	Conflicts between tenants and owners are usually handled informally and may remain unresolved.
Fragmented System	Users need separate channels for rent, buy, sell, sublet, bachelor housing, and agency support.

Table 3: Major Problems in the Current Real Estate Market

These issues are serious because housing decisions involve money, safety, family needs, and long-term planning. A family may rent first and later want to buy a home, but there is no single trusted system that supports the full journey. Students and job holders also face difficulty finding suitable bachelor rooms or roommates near their institutions or workplaces. Thikana addresses these problems through a unified, verified, and data-supported platform.

6. Motivation and Objectives

The main motivation behind Thikana is to build a complete property solution for Bangladesh where a user can move from renting to buying without changing platforms. The team was also motivated by the success of commission-based digital platforms such as ride-sharing and accommodation marketplaces, where technology reduces uncertainty and connects users directly.

Objective	Description
Simplify Property Search	Provide fast keyword, filter, and map-based property discovery.
Reduce Communication Barriers	Enable in-app chat and direct calling between users and property owners.
Build Trust	Use phone verification, NID simulation, reviews, ratings, and fake listing reports.
Improve Decision-Making	Provide area intelligence, commute estimation, 360-degree tours, and floor plan details.
Support Digital Transactions	Allow rent booking and payment through BDT-supported gateways such as SSLCommerz, bKash, or Nagad.
Support Future Growth	Build a scalable architecture that can expand across all divisions of Bangladesh.

Table 4: Project Objectives

7. Expected Features and Requirements

The following features define the functional scope of Thikana. The system is designed with both common listing features and unique smart features that differentiate it from a basic rental website.

Feature	Description
Smart Search	Keyword and filter-based instant property discovery for rent, buy, sell, sublet, and bachelor housing.
Advanced Filters	Filtering by price, location, room count, bathroom, balcony, furnished status, amenities, and property type.
Map-Based Location Search	Interactive map view for discovering nearby properties and understanding the surrounding area.
Image Gallery	Multi-photo upload and viewing for each listing, helping users inspect properties before visiting.

Save Favorites	Allows users to bookmark properties and receive availability alerts.
Direct Communication	In-app chat and direct calling between tenants, buyers, owners, and agencies.
AI Recommendations	Machine-learning based suggestions using budget, family size, workplace distance, preferred facilities, and user behavior.
Area Intelligence	Nearby schools, hospitals, markets, mosques, temples, traffic condition, safety rating, internet coverage, and local facility insights.
360-Degree Virtual Tour	Panoramic room views that allow users to inspect properties remotely before physical visits.
Floor Plan Visualization	2D floor plans with measurements and room annotations for better space understanding.
Tenant Verification	Student/job-holder verification, phone verification, NID simulation, and verified tenant badge.
Smart Agreement Generator	Digital rental agreements, tenant information sheets, and payment record support.
Rent and Expense Tracker	Monthly reminders, utility tracking, due alerts, and payment history.
Roommate Matching	Helps students and job holders find roommates near a selected area.
Owner Reputation System	Ratings, reviews, fake listing reports, and trust score for owners and agencies.
Analytics Dashboard	Shows listing views, enquiries, demand trends, and occupancy information for owners and agencies.

Table 5: Thikana - Core and Smart Feature Requirements

8. User Roles and System Modules

Thikana follows a role-based structure so that each type of user receives a dedicated interface and suitable permissions.

User Role	Capabilities
Tenant / Buyer	Browse listings, save favorites, chat with owners, book visits, view 360-degree tours, check area intelligence, and submit reviews.
Property Owner	Post and manage rent/sale listings, upload photos and floor plans, receive enquiries, set pricing, manage availability, and view analytics.
Real Estate Agency	Manage multiple properties, monitor tenants, handle bookings, and access agency-level reports.
Admin	Verify users, moderate listings and reviews, manage disputes, handle reports, suspend users, and monitor platform analytics.

Table 6: User Roles and Capabilities

Module	Function
Rental System	Family flat, bachelor room, and sublet listing management.
Real Estate Marketplace	Buy/sell property listing, property documents, agency listings, and booking support.
Smart Search and Recommendation	AI-powered discovery, filtering, price preference, and suggested listings.
Map and Location Intelligence	Area insights, nearby facilities, commute estimation, and map search.
Verification System	Phone verification, NID simulation, fake listing reports, and reputation system.
Analytics and Reports	Dashboard for owners, agencies, and admins showing trends, views, and user activity.

Table 7: Main System Modules

9. Development Approach

9.1 Development Model

Thikana will be developed using the Incremental Development Model. This model is suitable because the platform has many layers, and the team can deliver a working MVP first before adding advanced features. It also fits the academic sprint because each increment can be tested and corrected before the next increment begins.

Increment	Scope	Week
Increment 1: MVP	Property listing, smart search, user authentication, and basic map integration.	3-4
Increment 2: Core Features	Chat system, favorites, image upload, and phone verification.	5
Increment 3: Smart Experience	360-degree virtual tours, floor plans, and area intelligence.	6
Increment 4: AI and Analytics	AI recommendations, roommate matching, and analytics dashboard.	7
Increment 5: Final Integration	Payment integration, agreement generator, full testing, and deployment preparation.	8

Table 8: Increment Plan

9.2 System Workflow

A typical user flow begins with registration and verification. After login, the user searches by location, property type, budget, or keyword. The user can refine results using filters, check the map view, view photos, inspect 360-degree tours, check floor plans, and review area intelligence. After selecting a property, the user contacts the owner through chat or call, books a visit, and finalizes the deal through an agreement and payment record.

This workflow keeps the full journey inside one platform: discovery, comparison, communication, verification, and deal finalization.

10. Tools and Technology Stack

The selected tools are practical for a student team and also suitable for future scaling. The stack uses modern web development technologies, relational data storage, real-time services, maps, cloud storage, authentication, and payment support.

Component	Technology	Purpose
Frontend	React.js, HTML5, CSS3, Tailwind CSS	Component-based interface, responsive design, and modern styling.
Mobile Future Scope	Flutter or React Native	Cross-platform mobile application development.
Backend	Node.js, Express.js, or PHP Laravel	REST API, routing, business logic, and server-side operations.
Database	MySQL and Firebase	Structured relational data and real-time features.
Mapping	Google Maps API or Mapbox	Location services, map search, and proximity calculations.
Virtual Tour	Three.js or Panorama Viewer	360-degree room visualization and interactive walkthrough.
Payments	SSLCommerz, bKash, Nagad	BDT-based booking and rent payment integration.
Notifications	Firebase Cloud Messaging	Push alerts for enquiries, payment reminders, and availability updates.

Storage	Firebase or AWS Cloud Storage	Images, floor plans, documents, and virtual tour media.
Authentication	JWT and OAuth 2.0	Secure login, role-based access, and multi-factor support.
Design	Figma	Wireframes, UI prototypes, and design collaboration.
Version Control	Git and GitHub	Team collaboration, branch management, and version history.

Table 9: Technology Stack

11. Budget Estimation

The budget is divided into two parts. The first is a full project budget that estimates commercial-level development. The second is a student sprint budget for building the academic prototype within a short period.

11.1 Full Project Estimated Budget

Category	Estimated Cost (BDT)
UI/UX Design	1,00,000
Frontend Development	1,50,000
Backend and Database	2,00,000
360-Degree Tour and Visualization	80,000
AI and Smart Features	1,00,000
Cloud and Database for 6 Months	1,00,000
API Subscriptions for Maps and Payment	70,000
Testing and Quality Assurance	80,000
Deployment and Server Setup	50,000
Maintenance for First Year	1,00,000
Marketing and Miscellaneous	1,00,000
Total	11,30,000 BDT

Table 10: Full Project Budget

11.2 Student Initial Budget

Item	Cost (BDT)
Shared Hosting for 1.5 Months	2,000
Domain (.com) for 1 Year	1,500
Google Maps API Free Tier	0
Firebase Spark Plan	0
Figma Free Plan	0
Development Tools and Miscellaneous	8,500
Total Student Sprint Budget	12,000 BDT

Table 11: Student Initial Budget

12. Project Timeline

The proposed project timeline follows the Software Development Life Cycle with incremental delivery. Requirements and design come first, followed by MVP development, smart features, integration, testing, and deployment.

Phase	Duration	Description
Requirements Gathering	Week 1	Collect requirements, define scope, identify user roles, and finalize features.
UI/UX Design	Week 2	Prepare wireframes and prototypes in Figma.

Increment 1: MVP Development	Week 3-4	Build listing, search, authentication, and basic map features.
Increment 2: Core Features	Week 5	Add chat, favorites, image upload, and basic verification.
Increment 3: Virtual Tour and Intelligence	Week 6	Add 360-degree tour, floor plan, and area intelligence modules.
Increment 4: AI and Advanced Features	Week 7	Add AI recommendations, roommate matching, and analytics dashboard.
Integration and Testing	Week 8	Conduct system testing, bug fixing, and feature validation.
Deployment and Beta Launch	Week 8.5	Deploy the prototype and prepare final documentation.

Table 12: SDLC Timeline

13. Expected Outcomes

After successful development, Thikana is expected to provide measurable benefits to property seekers, owners, agencies, and administrators.

Outcome	Description
Faster Property Search	Users can discover suitable listings quickly through search, filters, and maps.
Improved Transparency	Verified listings, reviews, clear pricing, and commission information reduce uncertainty.
Better Decision-Making	Area intelligence, photos, floor plans, and virtual tours reduce unnecessary physical visits.
Trusted Ecosystem	Phone/NID verification, reporting tools, and reputation scores improve platform trust.
Centralized Solution	One platform supports rent, buy, sell, sublet, bachelor housing, and property management.
Scalable Foundation	The architecture can later expand to all eight divisions of Bangladesh and support mobile apps.

Table 13: Expected Outcomes

14. Conclusion

Thikana is designed as a complete digital real estate ecosystem for Bangladesh, not just a basic TO-LET application. It brings together rental listings, buy/sell support, owner dashboards, agency management, map-based search, verification, 360-degree virtual tours, AI recommendations, roommate matching, and analytics into one platform.

The project addresses real problems in the Bangladeshi property market: fake listings, broker dependency, time wastage, lack of verified information, poor location intelligence, and fragmented services. By following the Incremental Development Model, the team can deliver a working MVP first and then gradually add smart features based on feedback.

From an academic perspective, Thikana is a strong software engineering project because it includes requirement analysis, role-based modules, frontend and backend development, database design, API integration, security, testing, and deployment. Beyond the course, the concept has practical potential to become a useful product for tenants, buyers, owners, and agencies across Bangladesh.

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